Roll No. | | | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharmacy (Sem.-7)

INSTRUMENTAL METHODS OF ANALYSIS-THEORY

Subject Code : BP-701T

M.Code : 78387

Date of Examination : 20-11-2023

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly :**

   a) Define auxochrome. Give two examples.

   b) How does particle size of stationary phase correlate to resolution?

   c) What is electrophoretic mobility? How does it affect the migration rate of an analyte?

   d) Define overtone band.

   e) What is Rf value? Give an example.

   f) How fluorescence differs from phosphorescence?

   g) Why two monochromators are used in a spectrofluorimeter?

   h) Differentiate between normal phase and reverse phase chromatography.

   i) What are the different methods of excitation in atomic spectroscopy?

   j) What is a mulling agent?

SECTION-B

2. What is the principle of spectrofluorimetry? Discuss various factors affecting fluorescence intensity with examples.

3. Write a detailed account on principle, theoretical considerations, instrumentation, application and limitations of gas chromatography.

4. What are different modes of vibrations in an organic molecule? Describe various factors affecting vibrational frequencies in a molecule with examples.

SECTION-C

5. How does polarity of solvent affect the absorption maxima of an enone type compound? Support your answer with relevant examples.

6. Explain the working of a dispersive IR spectrophotometer with the help of a neat well labelled flow diagram.

7. What is the principle of turbidometry? How does a nephelometer differ from a turbidometer?

8. Discuss various sources of interferences in atomic emission spectroscopy.

9. Describe the principle and instrumentation of gel electrophoresis.

10. Explain the ion-exchange mechanism. Discuss about various factors affecting ion-exchange.

11. Write a note on various solute property detectors used in HPLC.

12. Give a comparative account of specific applications and limitations of paper, thin-layer and gel chromatographic techniques.

13. Describe the theoretical concepts behind affinity chromatography. Give its specific applications.

Chandigarh College of Pharmacy LIBRARY Landran (Mohali)

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

Dec-2023